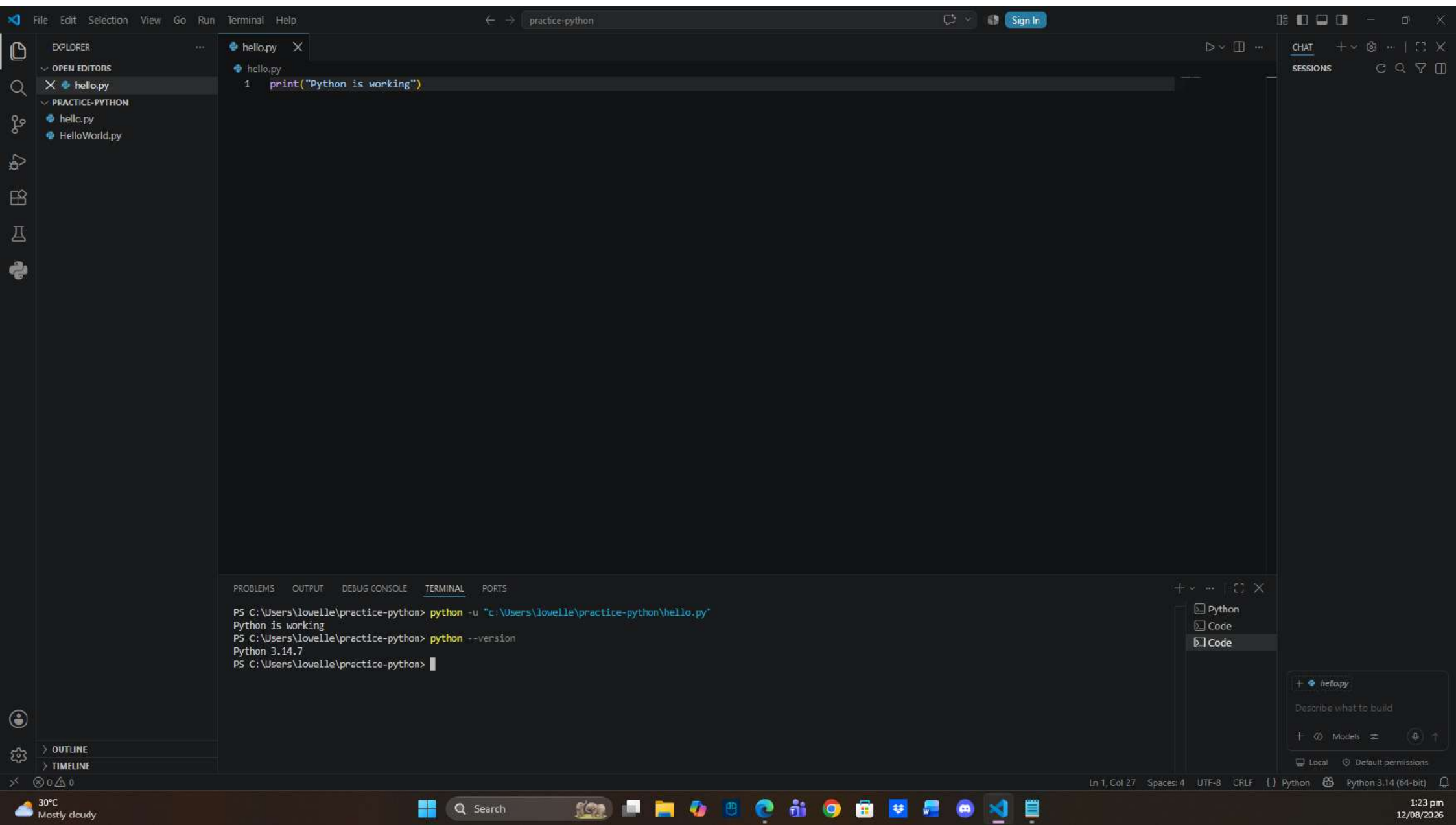


The terminal showing python --version and its answer.

VS Code with hello.py open and its output visible in the terminal.



The terminal showing the successful installation of NumPy, SciPy and Matplotlib.

The image shows a screenshot of a Visual Studio Code (VS Code) editor interface. The main editor window displays a Python script named `lab00_check_roallos.py`. The script contains two parts: a cross-check of a SciPy integration result and a Matplotlib plot. The cross-check calculates the integral of $\sin(x)$ from 0 to π and compares it to the exact value of 2. The Matplotlib part creates a plot with two lines: $\sin(x)$ (solid blue line) and $\cos(x)$ (dashed orange line). The plot is titled "Lab Assignment 0 - platform check" and is saved as `lab00_check.png`. The terminal window at the bottom shows the execution of the script, displaying the mean of $\sin(x)$ over one period as 0.0, the integral of $\sin(x)$ from 0 to π as 2.0, and the file path for the saved plot. On the right side, there is a chat interface with a "CHAT" tab and a "SESSIONS" list. The chat window shows a prompt to create an agent for a workflow, with a button to add a new agent. The bottom status bar indicates the current file is `lab00_check_roallos.py`, the editor is in "Python 3.14 (64-bit)" mode, and the workspace is named "lab00_check_roallos.py".

Visual Studio Code interface showing a Python project named "practice-python".

EXPLORER:

- OPEN EDITORS
 - hello.py
 - lab00_check_roallos.py
 - lab00_output_roallos.txt
- PRACTICE-PYTHON
 - figures
 - hello.py
 - HelloWorld.py
 - lab00_check_roallos.py
 - lab00_output_roallos.txt

lab00_output_roallos.txt

```
1 Python : 3.14.7
2 NumPy : 2.5.2
3 SciPy : 1.18.0
4 Matplotlib : 3.11.1
5
6 mean of sin(x) over one period : 0.0
7 integral of sin(x) from 0 to pi : 2.0 (exact value 2)
8
9 figure written to figures/lab00_check.png
10 PLATFORM READY.
11
```

TERMINAL:

```
PS C:\Users\lowelle\practice-python> python -u "c:\Users\lowelle\practice-python\hello.py"
Python is working
PS C:\Users\lowelle\practice-python> python --version
Python 3.14.7
PS C:\Users\lowelle\practice-python> python -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in C:\Program Files\Python314\Lib\site-packages (26.2.1)
PS C:\Users\lowelle\practice-python> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (2.5.2)
Requirement already satisfied: scipy in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (1.18.0)
Requirement already satisfied: matplotlib in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\lowelle\practice-python> python -u "c:\Users\lowelle\practice-python\lab00_check_roallos.py"
PS C:\Users\lowelle\practice-python> python lab00_check_<surname>.py | Out-File -Encoding utf8 lab00_output_<surname>.txt
Out-File : Illegal characters in path.
At line:1 char:35
+ ... eck_<surname>.py | Out-File -Encoding utf8 lab00_output_<surname>.txt
+ ~~~~~
+ CategoryInfo          : OpenError: (:) [Out-File], ArgumentException
+ FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\Users\lowelle\practice-python> python lab00_check_roallos.py | Out-File -Encoding utf8 lab00_output_roallos.txt
PS C:\Users\lowelle\practice-python> python lab00_check_roallos.py | Out-File -Encoding utf8 lab00_output_roallos.txt
PS C:\Users\lowelle\practice-python>
```

PROBLEMS:

- CategoryInfo : OpenError: (:) [Out-File], ArgumentException
- FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

STATUS BAR: Ln 11, Col 1 | Spaces: 4 | UTF-8 with BOM | CRLF | Plain Text

CHAT: Describe what to build

SESSIONS:

SYSTEM TRAY: Flood warning In effect | Search | Taskbar icons | 1:35 pm 12/08/2026

The full console output of your verification program, ending in PLATFORM READY.

The image shows a Windows 10 desktop with a dark-themed Visual Studio Code (VS Code) editor window open. The editor has three tabs: 'hello.py', 'lab00_check_roallos.py', and 'lab00_output_roallos.txt'. The 'lab00_output_roallos.txt' tab is active, displaying the output of a Python script. The output text is as follows:

```
1 Python : 3.14.7
2 NumPy : 2.5.2
3 SciPy : 1.10.0
4 Matplotlib : 3.11.1
5
6 mean of sin(x) over one period : 0.0
7 integral of sin(x) from 0 to pi : 2.0 (exact value 2)
8
9 figure written to figures/lab00_check.png
10 PLATFORM READY.
11
```

The left sidebar of VS Code shows the Explorer view with a file tree containing 'hello.py', 'lab00_check_roallos.py', and 'lab00_output_roallos.txt'. Below the Explorer is the 'PRACTICE-PYTHON' section with a list of files including 'figures', 'hello.py', 'HelloWorld.py', 'lab00_check_roallos.py', and 'lab00_output_roallos.txt'. The bottom of the VS Code window features a TERMINAL panel with a command prompt showing the execution of the script and the resulting error message: 'Out-File : Illegal characters in path.' The Windows taskbar at the bottom displays the system clock as 1:30 pm on 12/08/2026, along with various application icons and a search bar.

Your GitHub repository page, showing the repository name and the uploaded file.

Thanks for downloading Visual Studio | Lab Assignment 0 — Setting Up Your Environment | genelowelle-roallos/numerical-me

https://github.com/genelowelle-roallos/numerical-methods-lab

House design 12x1... House design 11x1... Why Fantasy Matte... courage without fe... who said without f... Quote by Christop... Integration by Sub... House Design Idea... Other favorites

genelowelle-roallos / numerical-methods-lab

Type / to search

Code Issues Pull requests Agents Actions Projects Security and quality Insights Settings

numerical-methods-lab

Private

Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

Code

genelowelle-roallos

Add files via upload

8ef6a4b · 22 minutes ago

4 Commits

README.md

Initial commit

14 hours ago

hello.py

Add files via upload

22 minutes ago

README

numerical-methods-lab

About

No description, website, or topics provided.

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Contributors 1

30°C Mostly cloudy

Search

ENG US

1:44 pm 12/08/2026

Include the figure figures/lab00_check.png

EXPLORER

OPEN EDITORS

lab00_check.png figures

PRACTICE-PYTHON

figures

lab00_check.png

hello.py

HelloWorld.py

lab00_check_roallos.py

lab00_output_roallos.txt

File Edit Selection View Go Run Terminal Help

practice-python

lab00_check.png

figures > lab00_check.png

lab00_check.png

Lab Assignment 0 - platform check

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Requirement already satisfied: pyparsing>=3 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\lowelle\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\lowelle\practice-python> python -u "c:\Users\lowelle\practice-python\lab00_check_roallos.py"
PS C:\Users\lowelle\practice-python> python lab00_check_<surname>.py | Out-File -Encoding utf8 lab00_output_<surname>.txt
Out-File : Illegal characters in path.
At line:1 char:35
+ ... eck_<surname>.py | Out-File -Encoding utf8 lab00_output_<surname>.txt
+ ~~~~~
+ CategoryInfo : OpenError: (:) [Out-File], ArgumentException
+ FullyQualifiedErrorId : FileOpenFailure,Microsoft.PowerShell.Commands.OutFileCommand

PS C:\Users\lowelle\practice-python> python lab00_check_roallos.py | Out-File -Encoding utf8 lab00_output_roallos.txt
PS C:\Users\lowelle\practice-python> python lab00_check_roallos.py | Out-File -Encoding utf8 lab00_output_roallos.txt

OUTLINE
TIMELINE

Python
Code
Code

lab00_check_roallos.py

Describe what to build

Local Default permissions

Whole Image 780x480 47.74KB

1:53 pm
12/08/2026